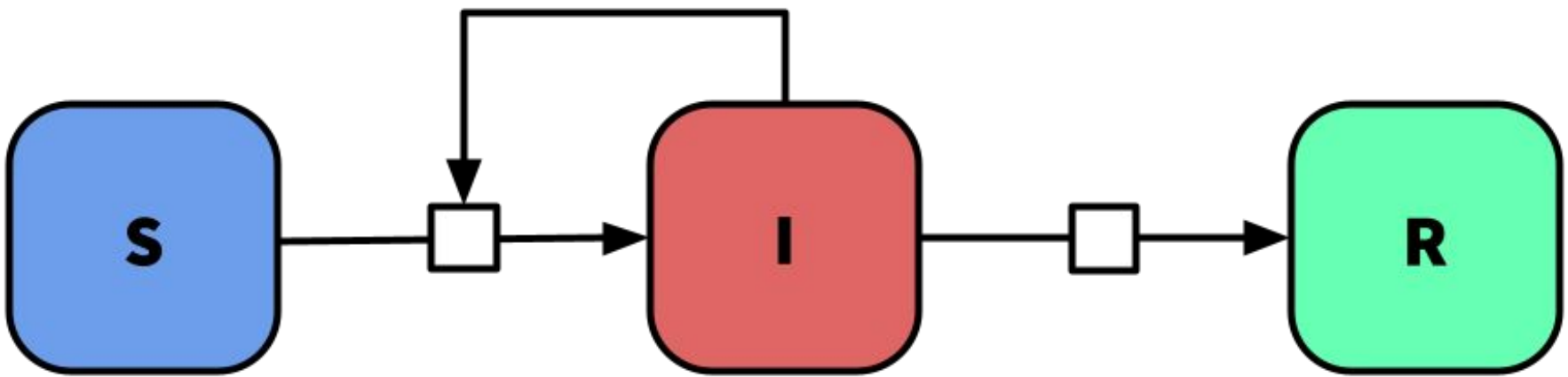


Probabilistic Cellular Automata for SIR Model

Justin Deckers
CSE 185

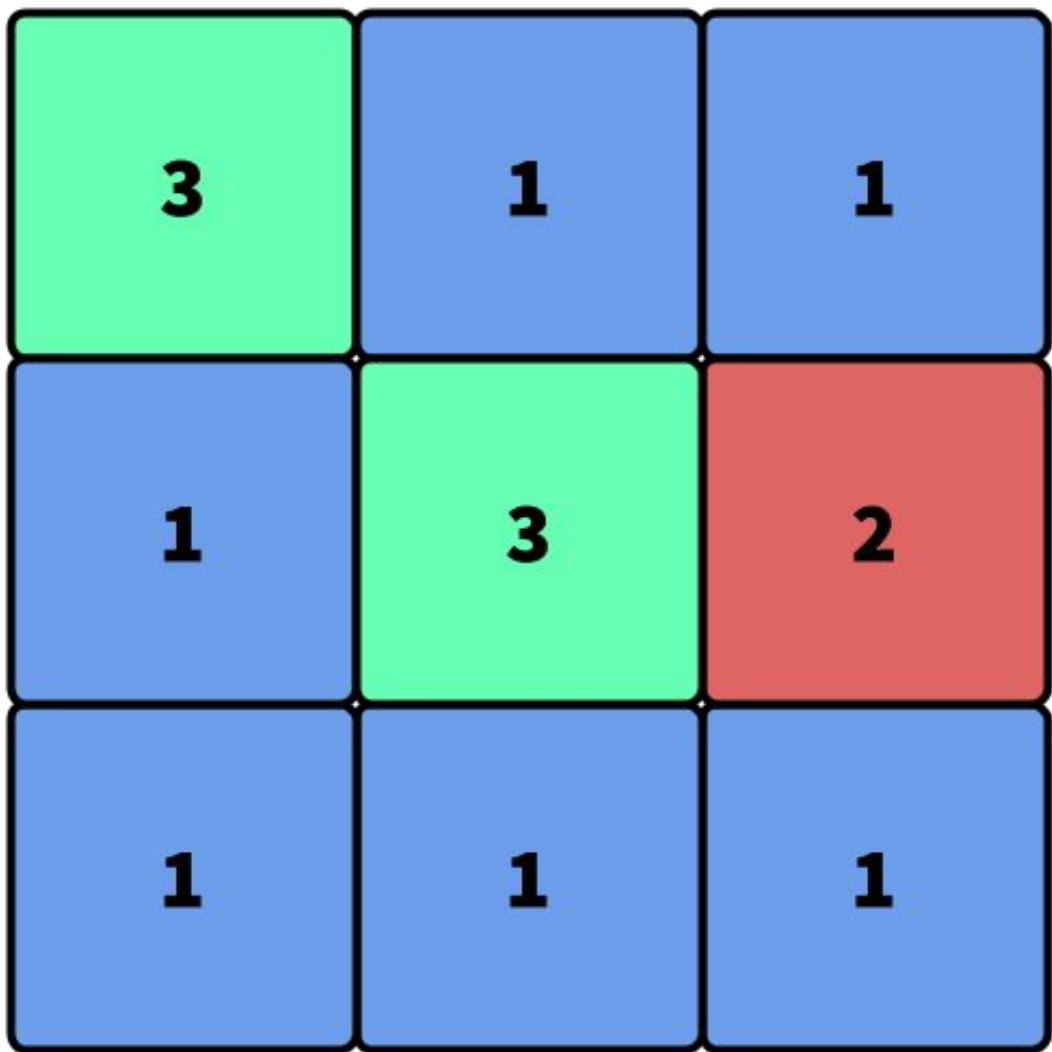
Rules for Disease Spread



- ❑ Susceptible individuals can become infected if they are in close contact with infected people
- ❑ After a while a Infected individual will recover and can no longer spread the virus

S: susceptible Individuals (0)
I: infected individuals (1)
R: recovered individuals (2)

The SIR model has 3 different states:
S- not exposed to virus yet
I- can spread the virus to susceptible people
R- can no longer be infected or spread the disease (assumed to have acquired immunity)



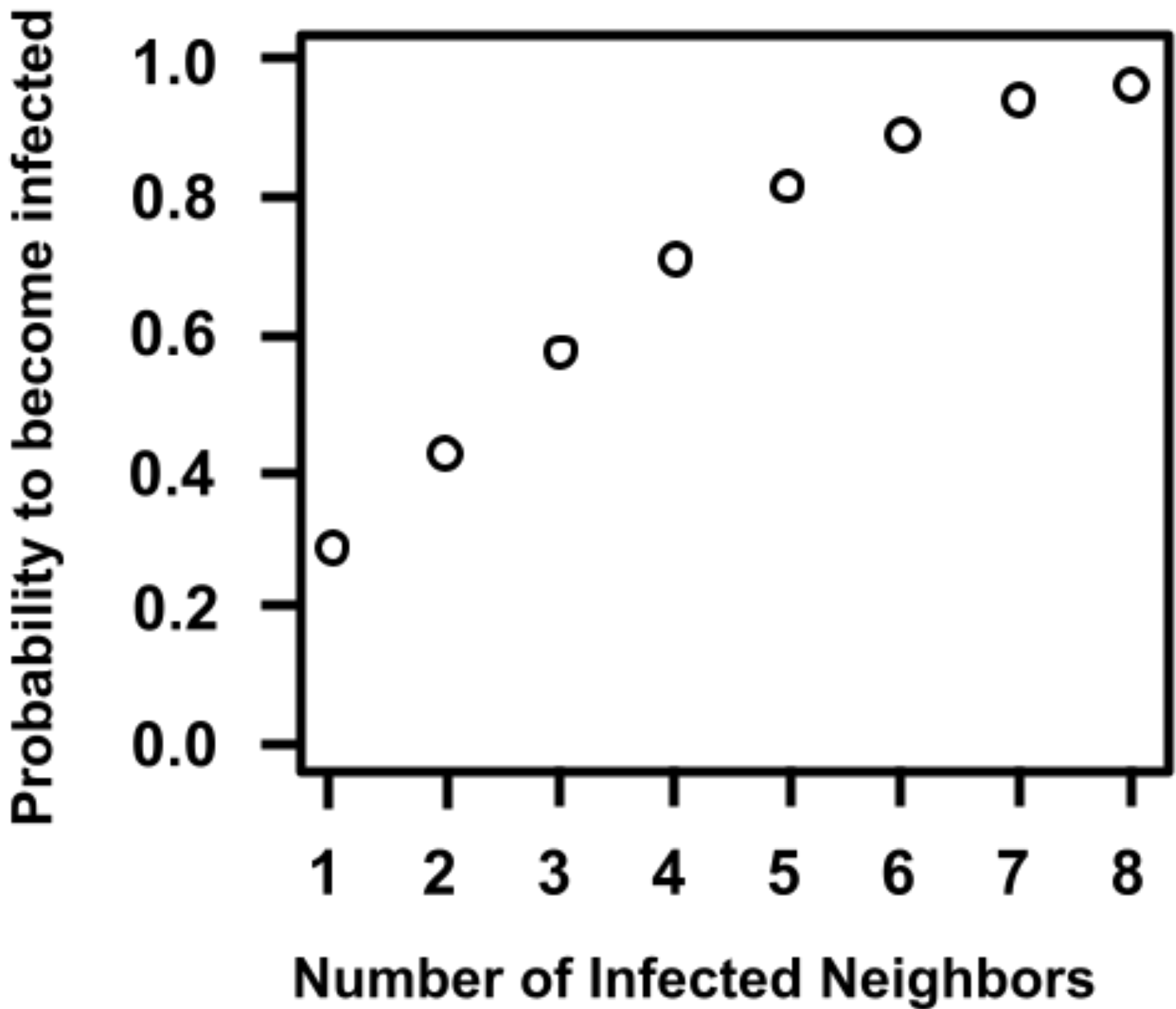
Each cell has 8 neighbors

Probability Function

$$P_{\text{infect}} = \frac{1}{1 + e^{-(a+bx)}}$$

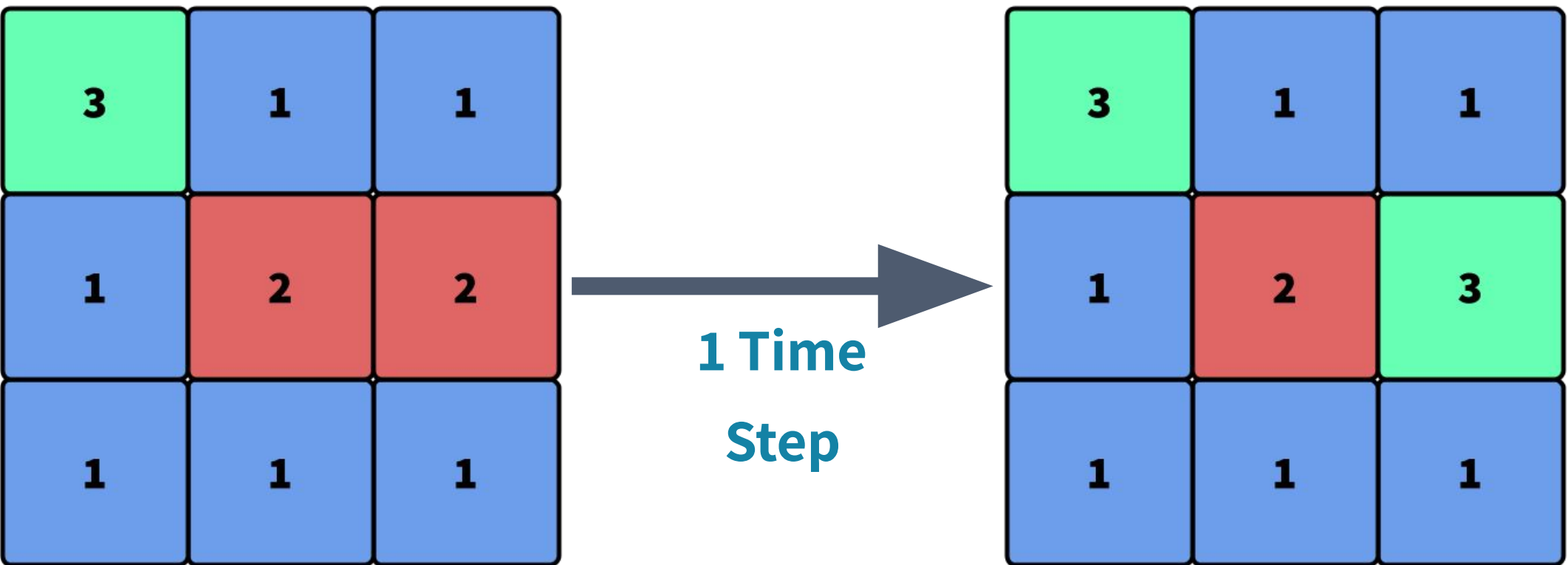
$b = 0.6$
 $a = -1.5$

x = Number of Infected Neighbors

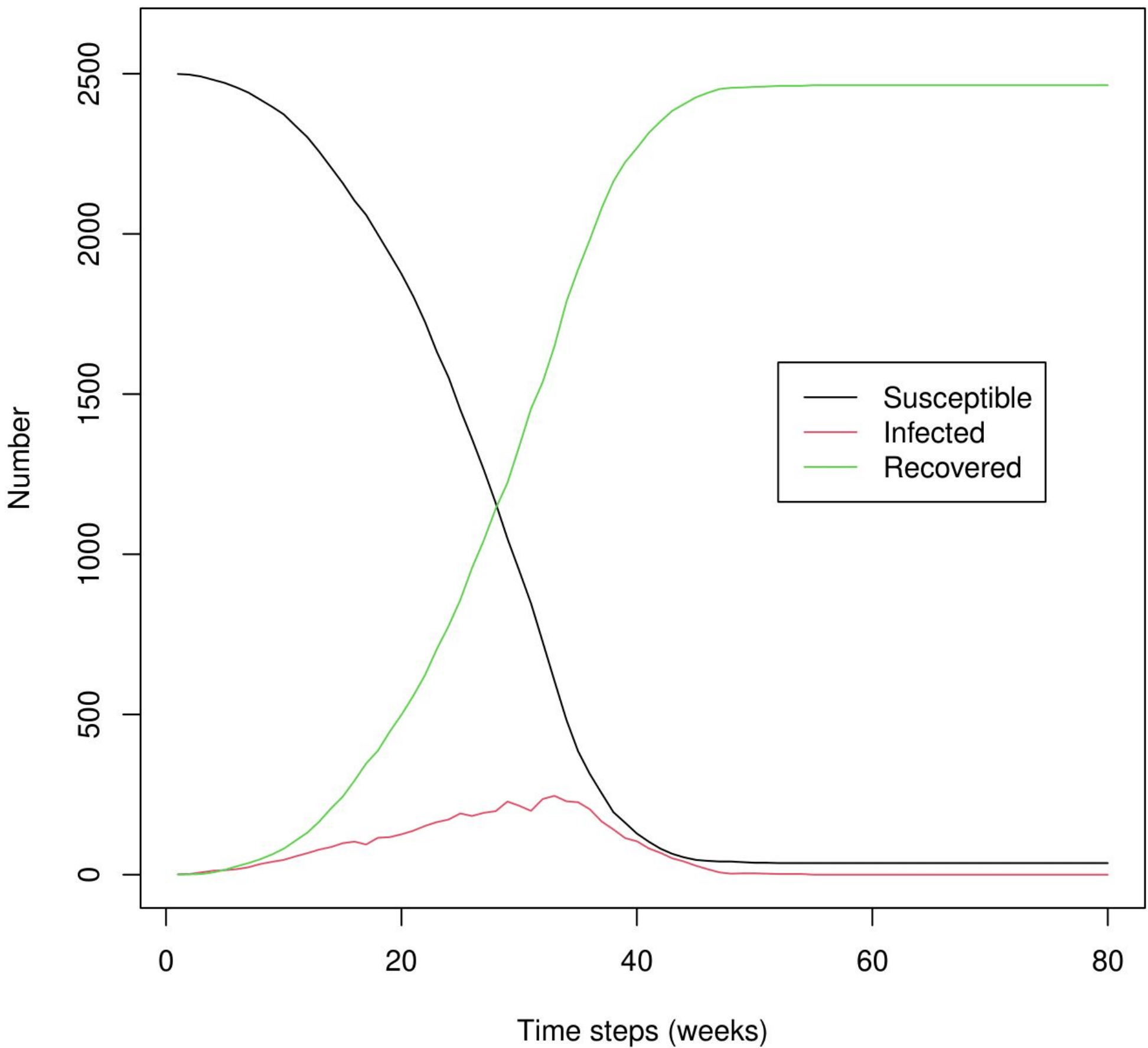


$$P_{\text{recover}} = 0.5 / \text{week}$$

∴ Half the population is expected to recover in one time step
Most individuals will take 1 - 3 weeks to recover



Conclusion



After running the script we can see the peak of the outbreak was around 35 weeks

Acknowledgements

- [1] S. Chang, "Cellular Automata Model for Epidemics." Available: <https://csc.ucdavis.edu/~chaos/courses/nlp/Projects2008/SharonChang/Report.pdf>
- [2] S. Ghosh and S. Bhattacharya, "Computational Model on COVID-19 Pandemic Using Probabilistic Cellular Automata," SN Computer Science, vol. 2, no. 3, Apr. 2021, doi: <https://doi.org/10.1007/s42979-021-00619-3>.